LEARNING MODULE

29

Convex Optimization in Practice

Limits on short sales, caps, leverage and trading turn mean–variance problems into convex programs: solve them, read their multipliers as prices, and check that the solver is right.

VOLUME V · Portfolio Optimization

BUILDS ON LM4 · LM7 · LM8 · LM9 · LM10 · LM11 · LM12 · LM27 · LM28

LEADS TO LM30 · LM31 · LM32 · LM33 · LM34 · LM35

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 prove that a convex problem's local minimum is global, and test convexity by Hessians, sums and maxima
☐	2 write portfolio limits as linear constraints, and recognize infeasible, unbounded and non-convex problems
☐	3 derive the KKT conditions of long-only minimum variance, and prove them sufficient for a convex objective with linear constraints
☐	4 compute R3 and R4 in their long-only form, and explain each zero weight by its multiplier
☐	5 compute the shadow price of a limit, and check it by moving the limit
☐	6 add trading costs to the objective, find the no-trade band, and weigh turnover against in-sample risk
☐	7 solve a problem in cvxpy with scaled inputs and set tolerances, and verify it by KKT residuals
☐	8 explain what KKT necessity needs and what a solver's status guarantees, and choose a solution method

LEARNING MODULE OVERVIEW

- On the through case's last window, LM28's tangency formula puts 114.24% in 0050 and shorts two funds. Forbidding short sales and borrowing lowers the in-sample Sharpe ratio only from 2.18 to 2.11.
- A convex problem has no false minima: every local minimum is global, (29.3). Portfolio variance is strictly convex because its Hessian 2Σ is positive definite.
- At long-only minimum variance every held fund has a beta of 1 to the portfolio and every excluded one a beta of at least 1, (29.5). On U28, 19 sectors are excluded, the closest at 1.000019.
- R4 holds a fund only when its excess return covers what its beta to the portfolio requires: 00646 earns 16.437 points a year above cash, but its beta requires 21.623.
- A multiplier is a rate: raising a 40% cap on R3 to 41% lowers its variance by 0.000102, against the linear prediction 0.000105.
- Trading costs in the objective create a no-trade band. At a cost weight of 1, the penalized path plans 31.5% of R3's turnover and leaves 31 of 91 signals untraded.
- On daily inputs CLARABEL's default tolerances miss R3's weights by up to 2.1×10^{-3}, on annual inputs by 3.4×10^{-5}. Every printed solution here is checked by its KKT residuals.

1 INTRODUCTION

LM28 solved mean–variance problems with equality constraints, in closed form. Real mandates add inequalities: no short sales, caps, limits on leverage and on trading. This module adds those limits, a solver that handles them, and the checks that show the solver is right.

MOTIVATING CASE **Why does the long-only portfolio drop two funds?**

On 30 September 2026 the through case re-estimates its inputs from its last 504 joint trading days, 23 August 2024 to 30 September 2026. Fed those means and that covariance matrix, LM28's tangency formula puts 114.24% in 0050. It sells short 27.75% of 00646 and 48.48% of 00679B, and it buys 31.07% of 00635U and 30.92% of 00642U. Its gross exposure is 252%, for an in-sample Sharpe ratio of 2.18 (Exhibit 1).

The formula shorts at every one of the through case's 92 month-ends. A reader with a cash account can neither short nor borrow. Forbid both, and the best Sharpe ratio falls only to 2.11. That portfolio holds 62.77% in 0050, 18.37% in 00635U and 18.86% in 00642U, and nothing in the other two.

Three questions follow. Why is 00646's weight zero, when it earns 16.44 points a year above cash? What does each limit cost? And when a solver prints these weights, how can the reader tell they are right?

Exhibit 1: The last window: inputs, the tangency portfolio and R4

Fund	Mean (%)	Volatility (%)	e_i (%)	Tangency (%)	R4 (%)
0050	48.78	26.32	47.28	114.24	62.77
00646	17.94	17.89	16.44	−27.75	0
00679B	−5.78	13.11	−7.28	−48.48	0
00635U	20.68	24.30	19.18	31.07	18.37
00642U	36.10	49.71	34.60	30.92	18.86
Sharpe ratio				2.18	2.11
Gross exposure (%)				252	100
Volatility (%)				31.94	18.85

Note: in-sample, on the 504 joint dates 2024-08-23 → 2026-09-30, annualized with $A = 242.95$; $e_i = \mu_i - r_f$, with $r_f = 1.5\%$ illustrative. The tangency is LM28's (28.6), not a rule.

Source: FinMind `TaiwanStockPrice`, `TaiwanStockDividendResult` and `TaiwanStockSplitPrice` (E5), retrieved 24 October 2026. Computations: `data/lm29_numbers.py`.

Sections 2 to 4 develop the theory, Sections 5 to 7 apply it to the through case, and Sections 8 and 9 solve and check in Python.

> **NOTE** **Notation for this module**
>
> E5 has $N = 5$ funds and U28 $N = 28$ sectors, indexed by i and j. In each window $\Sigma = A\hat{\Sigma}$ (divisor T) and $\boldsymbol{\mu} = A\bar{\mathbf{R}}$ are annual and treated as known; $\mathbf{e} = \boldsymbol{\mu} - r_f\mathbf{1}$. As in LM7, h, ψ_i, J and H are an objective, a constraint function, an optimal value and a Hessian; δ is LM28's risk aversion.
>
> Local: $\mathbf{u}$, $\mathbf{v}$, ω two points and the mixing weight of (29.1), not $\mathbf{x}$, $\mathbf{y}$, since $\mathbf{y}$ is R4's variable; $\mathcal{C}$ a convex set; x a real number; b_i the level of $\psi_i(\mathbf{w}) \le b_i$ or of a cap, not an entry of $\mathbf{b}$; $\boldsymbol{\lambda}$ the multipliers of $K\mathbf{w} = \mathbf{b}$ in (29.4); ν_i^{cap} a cap's multiplier; V the optimal variance of a capped problem (σ_p^2 is any portfolio's); $\mathbf{y}$ R4's homogenized variable; $\boldsymbol{\Delta}^{\pm}$ buys and sells, $\mathbf{w}^{\pm}$ long and short parts; θ the weight of a trading penalty; τ a holding period in years.

NOTE **Research hygiene**

Every portfolio here treats its window's means and covariance matrix as known, so its weights, Sharpe ratio and volatility are **in-sample** plug-in estimates with an unmeasured error. The windows are the through case's, fixed in advance. The caps, limits, spreads and cost weights are **constructed** illustrations, not rules, and none is tested on later data. R3's and R4's later results are LM30's, LM33's and LM34's.

Versions compared on the last window: four maximum-Sharpe portfolios, two relaxed bounds, one LP, one target on two frontiers and three statuses. On the same window R3 takes five cap levels: none, 39%, 40%, 41% and 42%. Elsewhere, seven cost weights, one turnover limit, two spreads, two other values of r_f and the widget's state. None is a candidate rule: the out-of-sample family stays R1–R8. Counts over the 92 signals are **descriptive**.

2 CONVEX SETS AND CONVEX FUNCTIONS

- [ ] prove that a convex problem's local minimum is global, and test convexity by Hessians, sums and maxima

A solver walks downhill, and on a bowl only the bottom can stop it. Convexity makes both the feasible set and the objective bowl-shaped.

DEFINITION **Convex set and convex function**

A set $\mathcal{C} \subseteq \mathbb{R}^N$ is **convex** if it contains the segment between any two of its points. A function h on a convex set $\mathcal{C}$ is **convex** if every chord lies on or above its graph. For all $\mathbf{u}, \mathbf{v} \in \mathcal{C}$ and $\omega \in [0,1]$:

$$\begin{aligned} &\omega\mathbf{u} + (1-\omega)\mathbf{v} \in \mathcal{C}, \\ &h\big(\omega\mathbf{u} + (1-\omega)\mathbf{v}\big) \\ &\leq \omega h(\mathbf{u}) + (1-\omega)h(\mathbf{v}). \end{aligned} \tag{29.1}$$

The function h is **strictly convex** if the inequality is strict whenever $\mathbf{u} \neq \mathbf{v}$ and $0 < \omega < 1$.

LM4's definition is the case $N = 1$, and LM10's (10.10) shows that a quadratic with a PSD matrix satisfies (29.1). The budget hyperplane, the long-only orthant $\mathbf{w} \geq \mathbf{0}$, a cap's half-space $w_i \leq b_i$ and the gross ball $\|\mathbf{w}\|_1 \leq 1.5$ are convex, and so is any intersection of convex sets.

THEOREM **Hessian test and gradient inequality**

Let h be twice differentiable on a convex open set $\mathcal{C}$. Then h is convex exactly when its Hessian $H(\mathbf{u})$ is PSD at every $\mathbf{u} \in \mathcal{C}$, and strictly convex if $H(\mathbf{u})$ is PD everywhere. A differentiable convex h satisfies, for all $\mathbf{u}, \mathbf{v} \in \mathcal{C}$,

$$h(\mathbf{v}) \geq h(\mathbf{u}) + \nabla h(\mathbf{u})^{\top}(\mathbf{v} - \mathbf{u}). \tag{29.2}$$

Proof. Inequality (29.1) involves only the points of one segment, so h is convex exactly when each restriction $\omega \mapsto h(\mathbf{u} + \omega(\mathbf{v} - \mathbf{u}))$ is. By the chain rule its second derivative is $(\mathbf{v} - \mathbf{u})^{\top} H(\cdot)(\mathbf{v} - \mathbf{u})$. LM4's one-variable test, stated there without proof, makes the restriction convex exactly when that derivative is ≥ 0. Since $\mathcal{C}$ is open, $\mathbf{v} - \mathbf{u}$ takes every direction, so the condition is H PSD everywhere; PD gives strict convexity. ■

DEEP DIVE **The gradient inequality** optional · click to expand

Let h be differentiable and convex, and apply (29.1) at $\mathbf{u} + \omega(\mathbf{v} - \mathbf{u})$, $0 < \omega \leq 1$:

$$\frac{h\big(\mathbf{u} + \omega(\mathbf{v} - \mathbf{u})\big) - h(\mathbf{u})}{\omega} \leq h(\mathbf{v}) - h(\mathbf{u}).$$

As $\omega \to 0$ the left side tends to $\nabla h(\mathbf{u})^{\top}(\mathbf{v} - \mathbf{u})$, which gives (29.2) (Boyd and Vandenberghe 2004, §3.1.3, p. 69). For a quadratic it is (10.9) with $\boldsymbol{\Delta}^{\top} M \boldsymbol{\Delta} \geq 0$.

A sum of convex functions with non-negative weights is convex, by (29.1) term by term, and so is a pointwise maximum (Boyd and Vandenberghe 2004, §3.2.3, p. 80). So $|x| = \max(x, -x)$, $\max(0, -x)$ and $\|\mathbf{w}\|_1$ are convex, though none has a derivative at 0.

EXAMPLE 1

Which portfolio sets are convex?

On the last window, show that long-only portfolios with a 40% cap, gross exposure at most 150% and volatility at most 25% form a convex set. Show that a volatility floor and "at most three funds" do not.

SOLUTION

The budget, the orthant and the caps are linear, and the gross ball is convex by the triangle inequality. The variance has the PD Hessian 2Σ, so it is strictly convex by (29.2), and the ellipsoid $\mathbf{w}^\top\Sigma\mathbf{w} \le (25\%)^2$ and the intersection are convex.

A floor $\mathbf{w}^\top\Sigma\mathbf{w} \ge 0.01$ alone holds $\mathbf{w}$ and $-\mathbf{w}$ but not their midpoint $\mathbf{0}$. A third each in 0050, 00646 and 00679B, and a third each in 00679B, 00635U and 00642U, average to a portfolio of all five funds.

Integer limits, such as a number of funds or round lots, are not convex, and this module omits them.

THEOREM **Convex problems have no false minima**

Let h and every ψ_i be convex on $\mathbb{R}^N$, and consider

$$\begin{aligned} &\min_{\mathbf{w}} \; h(\mathbf{w}) \quad \text{subject to} \\ &\psi_i(\mathbf{w}) \le b_i \text{ for each } i, \quad K\mathbf{w} = \mathbf{b}. \end{aligned} \tag{29.3}$$

Problem (29.3) has a convex feasible set, and every local minimum is a global minimum (Boyd and Vandenberghe 2004, §4.2.2, p. 138). A strictly convex h has at most one minimizer.

Proof. Each set $\{\psi_i \le b_i\}$ is convex by (29.1). If a local minimum $\mathbf{w}^*$ had a feasible $\mathbf{v}$ with $h(\mathbf{v}) < h(\mathbf{w}^*)$, the feasible points $\mathbf{w}^* + \omega(\mathbf{v} - \mathbf{w}^*)$, $0 < \omega \le 1$, would have lower values by (29.1), arbitrarily close to $\mathbf{w}^*$. Two minimizers of a strictly convex h would have a midpoint of lower value. ■

By (29.2), a feasible point where $\nabla h = \mathbf{0}$ is also a global minimum. Theorem (29.3) does not say that a minimum exists: each optimum here is proved by exhibiting a KKT point (Section 4).

導讀 **凸性讓局部極小值就是全域最小值**

凸集（convex set）包含任兩點之間的線段；凸函數（convex function）的弦不會低於圖形，也就是凸組合（convex combination）的函數值不超過兩端函數值的同樣加權平均。檢查方法有三種：海森矩陣（Hessian）處處半正定（positive semidefinite）、凸函數以非負權重相加、凸函數取最大值，例如絕對值與交易成本（transaction cost）。LM10 證明二次函數是凸函數若且唯若矩陣半正定，本節把同樣的結論推廣到一般函數。凸問題的局部極小值（local minimum）一定是全域最小值（global minimum），嚴格凸函數（strictly convex function）最多只有一個最小解，所以往下走的 solver 不會被假的谷底困住。

KNOWLEDGE CHECK 1

Show that LM27's cost of a trade, $\sum_i \left[0.1425\% \, |\Delta_i| + \text{tax}_i \max(0, -\Delta_i)\right]$ with $\text{tax}_i \geq 0$, is convex, using only sums and maxima. Why is "at most three funds" not convex?

ANSWER

Each $|\Delta_i| = \max(\Delta_i, -\Delta_i)$ and each $\max(0, -\Delta_i)$ is a maximum of linear functions, added with non-negative weights. Two allowed portfolios can average to one of five funds (Example 1).

3 QUADRATIC AND LINEAR PROGRAMS

- [ ] write portfolio limits as linear constraints, and recognize infeasible, unbounded and non-convex problems

Most portfolio limits are linear, and the variance is a convex quadratic.

DEFINITION **Quadratic and linear programs**

A **quadratic program** (QP) minimizes $\mathbf{w}^\top M \mathbf{w}$ plus a linear term, with M PSD, subject to linear equalities $K\mathbf{w} = \mathbf{b}$ and linear inequalities $\psi_i(\mathbf{w}) \leq b_i$. A **linear program** (LP) has $M = 0$ (Boyd and Vandenberghe 2004, (4.34), p. 152).

Each limit is a row: the budget a row of K, long-only $-w_i \leq 0$, a 40% cap $w_i \leq 0.40$, and "at least 10% in 00679B" $-w_3 \leq -0.10$. R3 with a cap is the QP

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to}$$
$$\mathbf{1}^\top \mathbf{w} = 1, \quad -w_i \leq 0, \quad w_i \leq 0.40.$$

A gross limit $\|\mathbf{w}\|_1 \leq 1.5$ becomes linear after a split (LM8's gross exposure): $\mathbf{w} = \mathbf{w}^+ - \mathbf{w}^-$, $\mathbf{w}^\pm \geq \mathbf{0}$ and $\mathbf{1}^\top(\mathbf{w}^+ + \mathbf{w}^-) \leq 1.5$. The positive and negative parts of $\mathbf{w}$ are one split, and any split has $\mathbf{1}^\top(\mathbf{w}^+ + \mathbf{w}^-) \geq \|\mathbf{w}\|_1$. A turnover limit splits the same way.

A limit on volatility, such as $\mathbf{w}^\top \Sigma \mathbf{w} \leq (25\%)^2$, is a second-order cone constraint. cvxpy sends such problems to a conic solver, and this module does not develop cone programming.

EXAMPLE 2

A linear program: the highest mean under caps

On the last window, maximize $\boldsymbol{\mu}^\top \mathbf{w}$ subject to $\mathbf{1}^\top \mathbf{w} = 1$ and $0 \leq w_i \leq 40\%$. The three highest means are 48.775% (0050), 36.096% (00642U) and 20.678% (00635U).

SOLUTION

The candidate $\mathbf{w}^*$ holds 40% in 0050 and 00642U and 20% in 00635U, a mean of $0.4 \times 48.775 + 0.4 \times 36.096 + 0.2 \times 20.678 = 38.08\%$. Both portfolios sum to 1, so

$$\boldsymbol{\mu}^\top \mathbf{w} - \boldsymbol{\mu}^\top \mathbf{w}^* = \sum_i (w_i - w_i^*)(\mu_i - 20.678\%).$$

Every term is ≤ 0: the top two funds have $w_i \leq 40\% = w_i^*$ and higher means, the third fund's term is 0, and the other two have $w_i \geq 0 = w_i^*$ and lower means. The means differ, so equality needs $\mathbf{w}^*$'s weights in four funds, and the budget fixes the fifth: the optimum is unique. A linear objective concentrates; only a risk term diversifies.

EXAMPLE 3

Infeasible, unbounded and not convex

On the last window, give cvxpy three problems: minimum variance with a 15% cap on each fund; the highest mean with only the budget; and a volatility floor $\mathbf{w}^\top \Sigma \mathbf{w} \geq 0.01$ alone.

SOLUTION

The caps sum to 75%: `infeasible`, with `prob.value` `inf` and `w.value` `None`. With only the budget, buying 0050 against a short in 00679B raises the mean without limit: `unbounded`, again with `prob.value` `inf` and `w.value` `None`. The floor alone is not convex: cvxpy raises a `DCPError`.

With the budget added, the floor would exclude nothing: every fully invested portfolio has a variance of at least R3's 0.0102, because R3's bounds do not bind (Section 4).

PITFALL **The status is part of the answer**

`infeasible`, `unbounded` and `optimal_inaccurate` are answers, and `Variable.value` is `None` when there is no solution; a `DCPError` means that cvxpy cannot verify that the problem, as written, is convex. Read the status before the weights.

KNOWLEDGE CHECK 2

Write "at most 40% in any fund, at least 10% in 00679B, gross exposure at most 150%" as linear rows. Why is a 15% cap on each of five funds infeasible, and why does $\mathbf{w} \geq \mathbf{0}$ bound the highest mean?

ANSWER

With $\mathbf{w} = \mathbf{w}^{+} - \mathbf{w}^{-}$ and $\mathbf{w}^{\pm} \geq \mathbf{0}$, the rows are $w_i \leq 0.40$, $-w_3 \leq -0.10$ and $\mathbf{1}^{\top}(\mathbf{w}^{+} + \mathbf{w}^{-}) \leq 1.5$. Five caps of 15% allow 75% in all, so the budget cannot hold. Long-only weights lie in $[0, 1]$, so no mean can exceed the highest fund mean.

4 KKT CONDITIONS FOR LONG-ONLY PORTFOLIOS

- ☐ derive the KKT conditions of long-only minimum variance, and prove them sufficient for a convex objective with linear constraints
- ☐ explain what KKT necessity needs and what a solver's status guarantees, and choose a solution method

LM7's (7.13) states the KKT conditions for a maximization, with an equality in the Lagrangian as $-\lambda(\psi - b)$ and a multiplier $\nu_i \geq 0$ for each inequality. Applied to $-h$, it gives the form for a minimization.

THEOREM KKT conditions for a minimization

Let h and every ψ_i be differentiable. Minimize $h(\mathbf{w})$ subject to $\psi_i(\mathbf{w}) \leq b_i$ for each i and $K\mathbf{w} = \mathbf{b}$. A point $\mathbf{w}^{*}$ with multipliers $\boldsymbol{\lambda}$ and ν_i is a **KKT point** if

$$\begin{aligned} \nabla h(\mathbf{w}^{*}) &= K^{\top}\boldsymbol{\lambda} - \sum_i \nu_i \nabla \psi_i(\mathbf{w}^{*}), \\ K\mathbf{w}^{*} &= \mathbf{b}, \qquad \psi_i(\mathbf{w}^{*}) \leq b_i, \\ \nu_i &\geq 0, \qquad \nu_i\big(b_i - \psi_i(\mathbf{w}^{*})\big) = 0. \end{aligned} \tag{29.4}$$

If h is convex and every ψ_i is linear, a KKT point is a global minimum.

A bound $w_i \geq 0$, written $-w_i \leq 0$, enters stationarity as $+\nu_i$, and a cap $w_i \leq b_i$ as $-\nu_i^{\text{cap}}$.

Proof of sufficiency. Let $\mathbf{w}$ be feasible. By (29.2) and stationarity,

$$\begin{aligned} h(\mathbf{w}) \geq h(\mathbf{w}^{*}) &+ \boldsymbol{\lambda}^{\top} K(\mathbf{w} - \mathbf{w}^{*}) \\ &- \sum_i \nu_i \nabla \psi_i(\mathbf{w}^{*})^{\top}(\mathbf{w} - \mathbf{w}^{*}). \end{aligned}$$

Here $K(\mathbf{w}-\mathbf{w}^*)=\mathbf{0}$, and for linear ψ_i the last product is $\psi_i(\mathbf{w})-\psi_i(\mathbf{w}^*)$. Complementary slackness turns $\nu_i\psi_i(\mathbf{w}^*)$ into $\nu_i b_i$, so each term is $\nu_i(\psi_i(\mathbf{w})-b_i)\le 0$, and $h(\mathbf{w})\ge h(\mathbf{w}^*)$. ■

Boyd and Vandenberghe (2004, §5.5.3, p. 244) state sufficiency for every convex problem (29.3). The converse, that every minimum is a KKT point, needs a **constraint qualification**. For convex problems they state it under Slater's condition (p. 244), which for linear constraints is feasibility alone (p. 227). The converse is stated here without proof. LM7's independent binding gradients are another qualification; in a non-convex problem a KKT point need not be a minimum.

So, for a convex objective with linear constraints, the optima are exactly the KKT points. In Example 2, as $\min -\boldsymbol{\mu}^\top\mathbf{w}$, the budget's multiplier is $\lambda=-20.678\%$; the caps' multipliers $\mu_i-20.678\%$ are 28.097 and 15.418 points, and the bounds' $20.678\%-\mu_i$ are 2.741 and 26.454, all non-negative.

Long-only minimum variance. Minimize $\mathbf{w}^\top\Sigma\mathbf{w}$ subject to $\mathbf{1}^\top\mathbf{w}=1$ and $\mathbf{w}\ge\mathbf{0}$. Stationarity is $2\Sigma\mathbf{w}^*=\lambda\mathbf{1}+\boldsymbol{\nu}$. Multiplying by $\mathbf{w}^{*\top}$, the budget leaves λ and complementary slackness removes $\boldsymbol{\nu}$, so $\lambda=2\sigma_p^2$. With (27.5), $(\Sigma\mathbf{w}^*)_i=\operatorname{Cov}(R_i,R_p)$:

$$\begin{aligned} 2\Sigma\mathbf{w}^*=\lambda\mathbf{1}+\boldsymbol{\nu},\quad \boldsymbol{\nu}\ge\mathbf{0},\quad \nu_i w_i^*=0,\\ \lambda=2\sigma_p^2,\qquad \nu_i=2\big[\operatorname{Cov}(R_i,R_p)-\sigma_p^2\big],\\ \beta_{i,p}=1+\nu_i/\lambda. \end{aligned} \tag{29.5}$$

Every held fund has $\beta_{i,p}=1$ and every excluded fund $\beta_{i,p}\ge 1$. A little more of a held fund leaves the variance unchanged to first order; an excluded fund would raise it. Constraining weights also acts as a shrinkage of $\hat\Sigma$, which LM30 studies (Jagannathan and Ma 2003).

On E5, R3's bounds never bind: over the 92 signals its smallest weight is 0.13% (0050, 2024-11-29), so R3 equals LM28's $\mathbf{w}^{\mathrm{mv}}$ at every signal. An active set needs a larger universe.

EXAMPLE 4

Long-only minimum variance of U28

Solve (29.5) for U28's 28 sector price indices on the full sample ($T=4{,}382$, $A=245.48$; covariance only), and check the excluded sectors.

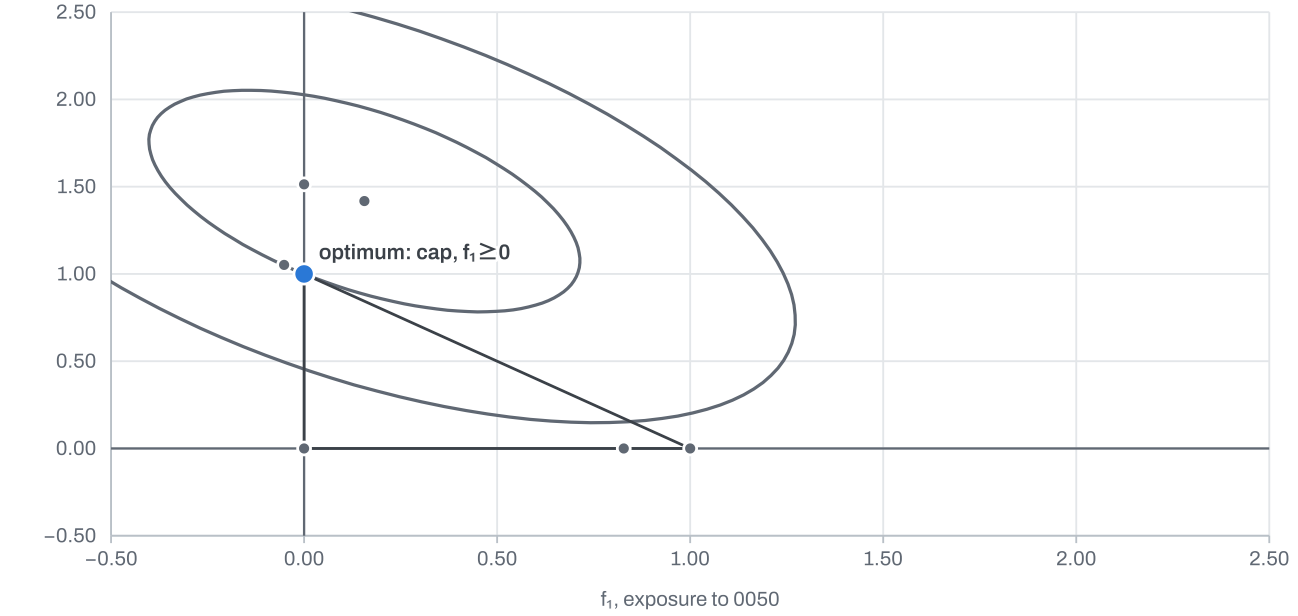
SOLUTION

Nine sectors are held, with an in-sample volatility of 12.10% and $\lambda=0.0293$. The 19 excluded sectors have $\beta_{i,p}$ from 1.000019 (Cement) to 1.480 (Optoelectronic), each ≥ 1. LM28's unconstrained $\mathbf{w}^{\mathrm{mv}}$ holds 12 shorts, with a gross exposure of 2.24. On the first half alone, the long-only portfolio again holds 9 sectors (LM30), only 7 of them the full sample's: the active set is itself an estimate.

U28's 28 bounds give $2^{28}=268{,}435{,}456$ candidate active sets, too many to enumerate as LM7 enumerated its eight. SciPy's `nnls`, the exact solver of Section 5, is an active-set method that solves the KKT conditions of non-negative least squares (Lawson and Hanson 1995). The widget tries the eight binding sets of a two-fund example. With all three constraints binding, the system is singular by (9.6), since three gradients in $\mathbb{R}^2$ are dependent.

Active sets with two funds and cash

INTERACTIVE



Binding	f_1	f_2	Multipliers	Feasible	Signs	Utility
none	0.16	1.42	—	no	yes	0.0652
$f_2 \geq 0$	0.83	0.00	−0.0567	yes	no	0.0250
$f_1 \geq 0$	0.00	1.51	−0.0081	no	no	0.0645
$f_1 \geq 0$, $f_2 \geq 0$	0.00	0.00	−0.0604, −0.0853	yes	no	0.0000
cap	−0.05	1.05	0.0278	no	yes	0.0572
cap, $f_2 \geq 0$	1.00	0.00	−0.0125, −0.0633	yes	no	0.0239
cap, $f_1 \geq 0$	0.00	1.00	0.0289, 0.0031	yes	yes	0.0571

Of 7 solvable binding sets, 4 pass feasibility and 3 the signs; only the optimum (cap, $f_1 \geq 0$ binding, utility 0.0571) passes both. Unconstrained point: (0.16, 1.42).

Model: maximize $f^T e - (\delta/2) f^T \Sigma f$ for 0050 (f_1) and 00646 (f_2), the rest in cash at the risk-free rate, subject to $f_1 + f_2 \leq$ cap, $f_1 \geq 0$ and $f_2 \geq 0$. Inputs: first window, 2017-02-13 → 2019-02-27, on its 504 joint dates; in-sample. Excess returns 6.04% and 8.53%, volatilities 13.50% and 11.87%, correlation 0.539. Each binding set solves stationarity with its constraints as equalities; the set with all three is singular. Ellipses: contours of the objective about the unconstrained point.

導讀 **KKT 條件：誰被排除、為什麼**

只作多（long-only）的最小變異數問題，每個權重下限 $w_i \geq 0$ 只有兩種狀態：鬆弛，乘數（multiplier）為 0；緊，乘數 $\nu_i \geq 0$，也就是有效限制（active constraint）。KKT 條件說，被持有的基金對投資組合的 beta 都等於 1，被排除的都至少是 1；互補鬆弛性（complementary slackness）保證乘數與權重至少一個是 0。乘數就是影子價格（shadow price）：放寬那條限制時，變異數下降的速率。對凸問題，KKT 條件是充分條件；限制條件（constraint）都是線性時，約束規範（constraint qualification）自動成立，所以也是必要條件。"active set" 本書保留英文。

KNOWLEDGE CHECK 3

At U28's long-only minimum variance, Cement is excluded with $\beta_{i,p} = 1.000019$, the closest of the 19 to 1. What do (29.5)'s conditions say, and how far would its covariance with the portfolio have to fall for Cement to enter? Why must R3 satisfy (29.5)?

ANSWER

Cement's $\nu_i = \lambda(\beta_{i,p} - 1)$ is tiny but positive: adding Cement would raise the variance, barely. Cement would enter once its covariance fell below σ_p^2, a fall of about 0.0019% of it, since then $\beta_{i,p} < 1$. R3 satisfies (29.5) because linear constraints satisfy a constraint qualification.

5 R3 AND R4 IN THE THROUGH CASE

- [ ] compute R3 and R4 in their long-only form, and explain each zero weight by its multiplier
- [ ] compute the shadow price of a limit, and check it by moving the limit

R3 is (29.5) on each window's Σ, solved exactly by `vol5_rules.py` (a Cholesky factor and `nnls`) and checked by its KKT conditions. R4 maximizes a ratio, which a change of variables turns into a QP.

The Sharpe ratio $\mathbf{e}^\top\mathbf{w}/\sqrt{\mathbf{w}^\top\Sigma\mathbf{w}}$ does not change when $\mathbf{w}$ is scaled by a positive number. Suppose some $e_i > 0$, so the best ratio is positive. For a long-only $\mathbf{w}$ with $\mathbf{e}^\top\mathbf{w} > 0$, set $\mathbf{y} = \mathbf{w}/\mathbf{e}^\top\mathbf{w}$; then $\mathbf{e}^\top\mathbf{y} = 1$ and the ratio is $1/\sqrt{\mathbf{y}^\top\Sigma\mathbf{y}}$. Maximizing it is minimizing $\mathbf{y}^\top\Sigma\mathbf{y}$, by Boyd and Vandenberghe's change of variables (2004, §4.3.2, p. 151):

$$\begin{gathered}\mathbf{y}^* \text{ minimizes } \mathbf{y}^\top \Sigma \mathbf{y} \\ \text{subject to } \mathbf{e}^\top \mathbf{y} = 1,\ \mathbf{y} \geq \mathbf{0}, \\ \mathbf{w}^* = \mathbf{y}^* / \mathbf{1}^\top \mathbf{y}^*, \\ \mathrm{SR}^* = \left(\mathbf{y}^{*\top} \Sigma \mathbf{y}^*\right)^{-1/2}, \qquad \lambda = 2/\mathrm{SR}^{*2}.\end{gathered} \tag{29.6}$$

Stationarity is $2\Sigma\mathbf{y}^* = \lambda\mathbf{e} + \boldsymbol{\nu}$, and multiplying by $\mathbf{y}^{*\top}$ gives $\lambda = 2\mathbf{y}^{*\top}\Sigma\mathbf{y}^*$. Substituting $\mathbf{y}^* = \mathbf{w}^*/(\mu_p - r_f)$, since $\mathbf{e}^\top\mathbf{w}^* = \mu_p - r_f$, and dividing by λ,

$$\begin{gathered}\frac{\nu_i}{\lambda} = \beta_{i,p}(\mu_p - r_f) - e_i \geq 0, \\ \text{with equality for held funds.}\end{gathered} \tag{29.7}$$

A held fund earns exactly its beta to the portfolio times the portfolio's excess return; an excluded fund earns less, by ν_i/λ. A positive excess return is not enough.

EXAMPLE 5

R4's multipliers at the last signal

Compute R4's KKT quantities at the last signal, and answer the case's first question.

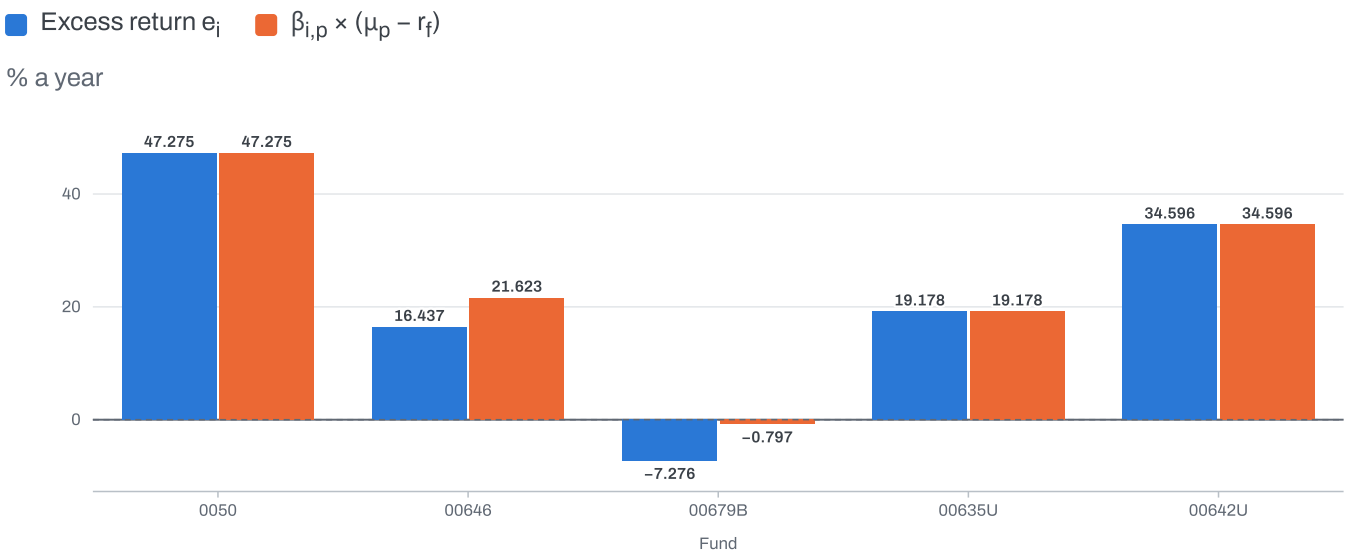
SOLUTION

The multiplier is $\lambda = 0.45062$ and $\mu_p - r_f = 39.72\%$:

	0050	00646	00679B	00635U	00642U
$\beta_{i,p}$	1.190	0.544	−0.020	0.483	0.871
ν_i	0	0.02337	0.02920	0	0
ν_i/λ (points)	0	5.19	6.48	0	0

00646 earns 16.437 points above cash, but its beta requires $0.54436 \times 39.722 = 21.623$ (Exhibit 2). Over the 92 signals, R4 holds one, two, three, four or five funds at 2, 21, 56, 13 and 0 signals. A fund with $e_i > 0$ was excluded in 80 of 338 cases. A fund with $e_i \leq 0$ was held 6 times, as a hedge with $\beta_{i,p} < 0$: 00679B on 2021-08-31, for example, at 14.4% with $e_i = -1.74\%$.

Exhibit 2: R4 at the last signal: what each fund earns and what its beta requires



Note: % a year; each fund's e_i beside $\beta_{i,p}(\mu_p - r_f)$, (29.7). The bars are equal for the held funds, and 00646 and 00679B earn 5.19 and 6.48 points less, ν_i/λ. In-sample, on the last window's 504 joint dates. Source: as Exhibit 1.

R4 needs one positive e_i, else it holds R3, and it exists even in the 22 windows where LM28's tangency does not. The rate $r_f = 1.5\%$ is illustrative. At 0% and 2%, R4 holds the same three funds, with no weight moving more than 1.15 points and Sharpe ratios of 2.19 and 2.08.

A multiplier also prices its limit. Let $J(b_i)$ be the optimal value as one level b_i moves.

RESULT **Multipliers are shadow prices**

For a convex problem (29.3) with linear constraints and KKT multipliers ν_i,

$$\frac{\partial J}{\partial b_i} = -\nu_i \text{ where } J \text{ is differentiable, and } J(b_i + \Delta b_i) \geq J(b_i) - \nu_i \, \Delta b_i \text{ for every } \Delta b_i. \tag{29.8}$$

Stated without proof. The derivative is Boyd and Vandenberghe's local sensitivity (2004, §5.6.3, p. 251); the bound is their global inequality (5.57), p. 250, valid by the refined Slater condition (p. 227).

EXAMPLE 6

A shadow price checked by moving the bound

In R4's QP (29.6) on the last window, relax $y_i \geq 0$ to $y_i \geq -0.01$, for 00646 and then for 00679B, and compare each gain with (29.8).

SOLUTION

Here $J = \mathbf{y}^{*\top}\Sigma\mathbf{y}^*$ and $\mathrm{SR} = 1/\sqrt{J} = 2.1067$. By (29.8), relaxing a bound by 0.01 lowers J by at most $\nu_i \times 0.01$, which would raise the ratio by about $\mathrm{SR}^3\nu_i \times 0.01/2$. For 00646, J falls by 0.000232 against $\nu_i \times 0.01 = 0.000234$, and the ratio rises by 0.001084, from 2.106733 to 2.107817, against 0.001093. For 00679B, J falls by 0.000290 against 0.000292, and the ratio rises to 2.108091, by 0.001358 against 0.001365. Inequality (29.8) bounds the fall of J, not the ratio's gain.

導讀 **正的超額報酬還不夠**

在貫穿案例（through case）的 92 個月底，切點投資組合（tangency portfolio）的公式每次都有放空；禁止放空與借款之後，最大夏普比率（Sharpe ratio）的問題可以改寫成二次規劃（quadratic programming）：令 $\mathbf{y} = \mathbf{w}/\mathbf{e}^\top\mathbf{w}$，最大化比率就等於在 $\mathbf{e}^\top\mathbf{y} = 1$ 下最小化 $\mathbf{y}^\top\Sigma\mathbf{y}$。KKT 條件說，被持有的基金，超額報酬（excess return）剛好等於它對投資組合的 beta 乘上投資組合的超額報酬；被排除的基金賺得比這個少。00646 每年超過無風險利率（risk-free rate）16.44 個百分點，但它的 beta 要求 21.62，所以權重是 0。乘數（multiplier）同時是影子價格（shadow price）：放寬一條限制，目標值的改善不會超過乘數乘上放寬的量。

KNOWLEDGE CHECK 4

From $\beta_{i,p} = 0.54436$, $\mu_p - r_f = 39.722\%$ and $e_i = 16.437\%$, find 00646's shortfall at R4. What would have to change for 00646 to be held?

ANSWER

The shortfall is $0.54436 \times 39.722 - 16.437 = 5.19$ points (5.18612 from unrounded inputs). Its excess return would have to rise by more than that, or its beta to the portfolio fall below $16.437/39.722 = 0.414$, so that its beta requires less than it earns.

6 CAPS, GROSS LIMITS AND THE LONG-ONLY FRONTIER

- ☐ compute the shadow price of a limit, and check it by moving the limit
- ☐ write portfolio limits as linear constraints, and recognize infeasible, unbounded and non-convex problems

On the last window R3 puts 56.5% in 00679B, so a 40% cap binds.

EXAMPLE 7

R3 with a 40% cap

Solve R3 on the last window with $w_i \le 40\%$, find the multipliers and the betas, and check the cap's price by raising it to 41%.

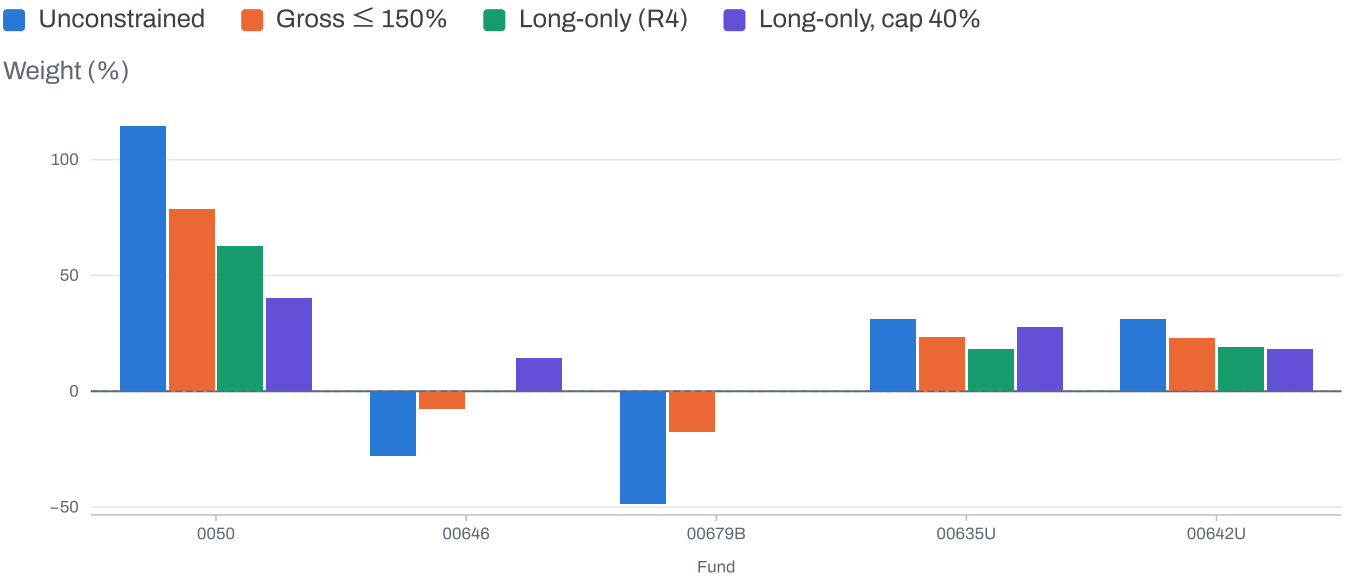
SOLUTION

The capped portfolio holds 6.0, 26.9, 40.0, 19.3 and 7.8%, a volatility of 10.52% against R3's 10.10%. Only 00679B's cap binds, so stationarity puts $(2\Sigma\mathbf{w})_i = \lambda$ for the free funds and $\lambda - \nu^{\text{cap}}$ for 00679B. Weighting and summing gives $\lambda = 2V + 0.40\,\nu^{\text{cap}}$, with $V = 0.0110665$, $\nu^{\text{cap}} = 0.010543$ and $\lambda = 0.026350$.

The free funds have $\beta_{i,p} = \lambda/2V = 1.191$, and 00679B $(\lambda - \nu^{\text{cap}})/2V = 0.714$. By (29.8) a 41% cap should change the variance by $-\nu^{\text{cap}} \times 0.01 = -0.000105$; the exact change is -0.000102. R3's largest weight exceeds 40% at 63 of the 92 signals.

Exhibit 3 prices limits in Sharpe ratio. It solves (29.6) on the last window with $\mathbf{y} \ge \mathbf{0}$ replaced by limits homogeneous in $\mathbf{y}$: none; $\|\mathbf{y}\|_1 \le 1.5\,\mathbf{1}^\top\mathbf{y}$; $\mathbf{y} \ge \mathbf{0}$ itself; and $\mathbf{y} \ge \mathbf{0}$ with $\mathbf{y} \le 0.40\,\mathbf{1}^\top\mathbf{y}$. The ratio falls from 2.18 to 2.16 with gross exposure at most 150%, 2.11 long-only (R4) and 2.00 with the cap: the case's second answer.

Exhibit 3: Maximum-Sharpe weights under four sets of limits



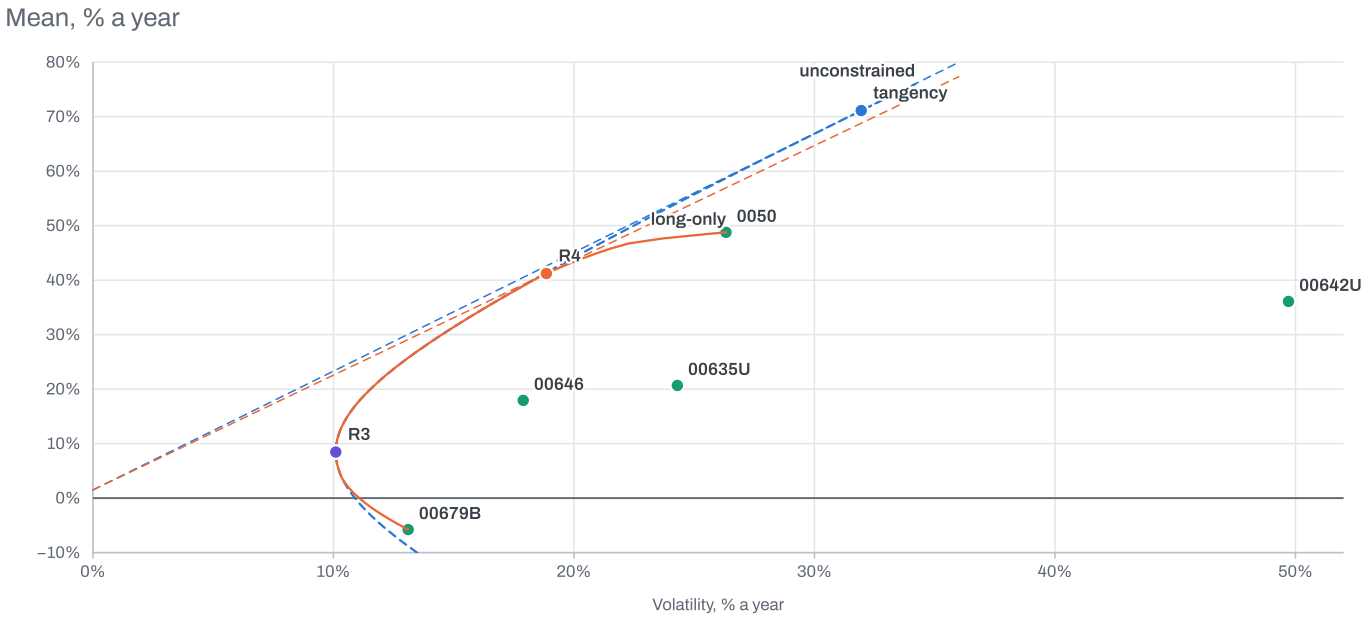
Note: weights in %; in-sample Sharpe ratios 2.18, 2.16, 2.11 and 2.00, volatilities 31.94%, 23.01%, 18.85% and 16.40%. Gross exposure at most 150% gives 78.82, −7.59, −17.41, 23.36 and 22.82% in Exhibit 1's order, and the 40% cap 40.00, 14.37, 0, 27.48 and 18.15%. The limits are constructed; "unconstrained" is LM28's tangency. In-sample, on the last window's 504 joint dates. Source: as Exhibit 1.

On Exhibit 4, the long-only and unconstrained frontiers coincide for target means from 4.12% to 30.44%, where LM28's frontier weights are all non-negative. Above that range the long-only frontier lies

to the right (at a target of 40%, 18.32% against 18.29%), since a smaller feasible set never gives a lower variance. It ends at 0050, the fund with the highest mean, 48.8%, at a volatility of 26.3%.

Minimizing a sample's expected shortfall instead of variance is a linear program (LM32).

Exhibit 4: The frontier with and without long-only limits



Note: % a year. The curves are LM28's unconstrained frontier (dashed) and the long-only frontier (solid), and the thin dashed lines are capital market lines from $r_f = 1.5\%$ with slopes 2.18 and 2.11. Points, as (volatility, mean): R3, the global minimum (10.10%, 8.46%); R4 (18.85%, 41.22%); the tangency (31.94%, 71.13%). In-sample, on the last window's 504 joint dates. Source: as Exhibit 1.

Financed leverage maximizes $\mathbf{f}^\top \mathbf{e}$ less a spread over r_f on borrowed money, under a volatility limit (Problem 11). If the gross limit is slack, a limit below R4's volatility gives R4 scaled to the limit, with $L = \mathbf{1}^\top \mathbf{f} < 1$ and the rest in cash. A limit above the volatility of the long-only tangency at the borrowing rate gives that tangency, scaled, with $L > 1$. In between, $L = 1$.

> **PITFALL** **A shadow price is a rate, not a total gain**
>
> Raising the cap from 40% to 41% lowers the variance by 0.000102, not the predicted 0.000105. For a convex problem the first-order estimate is always on the optimistic side, by (29.8).

KNOWLEDGE CHECK 5

With $\nu^{\text{cap}} = 0.01054$, predict the change in R3's variance when the cap rises from 40% to 41%. The exact change is -0.000102. Why does the prediction overstate the gain?

ANSWER

The prediction is $-0.01054 \times 0.01 = -0.000105$. By (29.8) the optimal value never falls below its tangent line, $J(b_i + \Delta b_i) \geq J(b_i) - \nu_i \, \Delta b_i$. As the cap rises, R3 moves weight back into 00679B, and each further point of cap is worth less.

7 COSTS IN THE OBJECTIVE

- [ ] add trading costs to the objective, find the no-trade band, and weigh turnover against in-sample risk

LM27's cost of a trade $\boldsymbol{\Delta}$, a fraction of pre-trade wealth, is 0.1425% of $\|\boldsymbol{\Delta}\|_1$ plus a 0.1% tax on sales, none for 00679B ((27.2); LM34 prints the sources). In the objective:

$$\begin{aligned} \min_{\mathbf{w}} \;\; & \mathbf{w}^\top \Sigma \mathbf{w} + \theta \, \text{cost}\left(\mathbf{w} - \mathbf{w}_t^{\text{drift}}\right) \\ \text{subject to} \;\; & \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}. \end{aligned} \tag{29.9}$$

The cost is a sum of maxima (Section 2), so for $\theta \geq 0$ the problem is convex. The weight θ is annual variance per unit of wealth paid once. In LM28's exposure form, a holding period of τ years trades $\frac{\delta\tau}{2}\mathbf{w}^\top\Sigma\mathbf{w}$ against a cost paid once, so $\theta = 2/(\delta\tau)$. Monthly trading, $\tau = 1/12$, with $\delta = 4$ gives $\theta = 6$.

Splitting each trade into buys and sells, $\mathbf{w} = \mathbf{w}_t^{\text{drift}} + \boldsymbol{\Delta}^+ - \boldsymbol{\Delta}^-$, makes the cost linear (Boyd and Vandenberghe 2004, p. 156). The problem is then a QP, and (29.4) gives, for a fund still held after the trade,

$$\begin{aligned} \text{bought: } (2\Sigma\mathbf{w})_i &= \lambda - \theta \, 0.1425\%, \\ \text{sold: } (2\Sigma\mathbf{w})_i &= \lambda + \theta \, (0.1425\% + \text{tax}_i). \end{aligned}$$

An untouched fund lies in the **no-trade band** between the two. cvxpy receives the cost as written, with `norm1` and `pos`.

EXAMPLE 8

One rebalance with costs

On 2025-04-30, R3's target of the 2025-03-31 signal, traded on 2025-04-01, has drifted. Compare a trade to the new R3 with the penalized trade at $\theta = 1$, planned at the signal close.

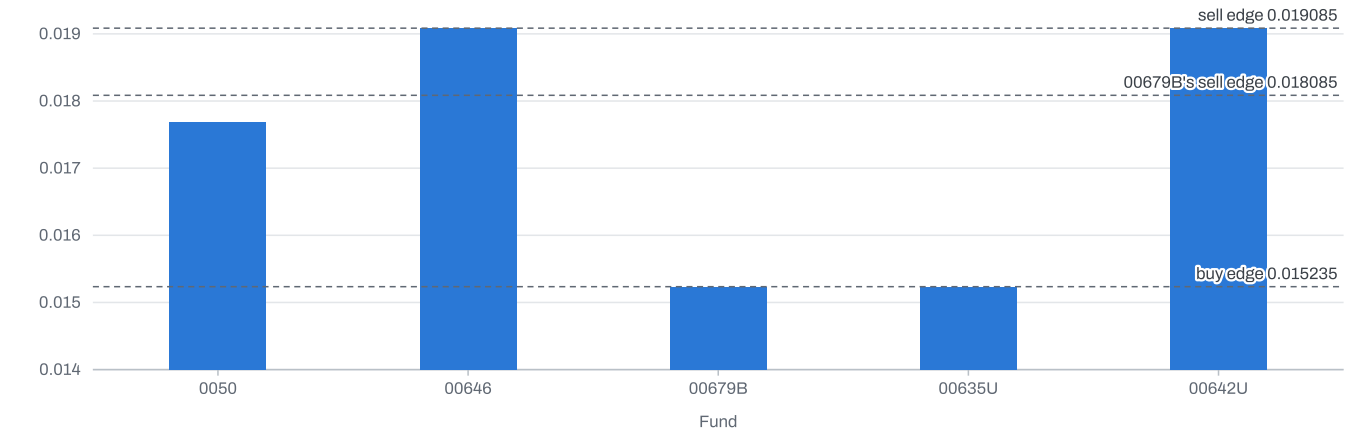
SOLUTION

	0050	00646	00679B	00635U	00642U
Drifted weight (%)	2.2	31.8	35.7	23.9	6.4
Trade at $\theta = 1$ (points)	0	−9.9	+4.6	+6.7	−1.4

Trading to the new R3 would have a turnover of 18.72% and cost 0.072% of wealth. The penalized trade has a turnover of 11.30% and costs 0.0435%, for an in-sample volatility of 9.03% against R3's 8.97% and no trade's 9.51%.

With $\lambda = 0.016660$ the band runs from 0.015235 (buy) to 0.019085 (sell), or 0.018085 for the untaxed 00679B (Exhibit 5). No trade is optimal for $\theta \geq 3.2022$, a threshold set by selling 00646 against buying 00635U.

Exhibit 5: The no-trade band on 2025-04-30 at $\theta = 1$



Note: the gradient $(2\Sigma\mathbf{w})_i$ at the penalized solution, with the band's three edges dashed; 0050 lies inside. In-sample, on the 2025-04-30 window's 504 joint dates. Source: as Exhibit 1.

DEEP DIVE **When not trading is optimal** optional · click to expand

Let every drifted weight be positive. Then $\mathbf{w} = \mathbf{w}_t^{\text{drift}}$ satisfies (29.9)'s KKT conditions exactly when one λ fits every band:

$$\begin{gathered}(2\Sigma\mathbf{w}_t^{\text{drift}})_i - \theta(0.1425\% + \text{tax}_i) \\ \leq \lambda \leq (2\Sigma\mathbf{w}_t^{\text{drift}})_j + \theta\, 0.1425\% \\ \text{for all } i, j.\end{gathered}$$

Such a λ exists exactly when every left side is at most every right side:

$$\theta \geq \max_{i \neq j} \frac{(2\Sigma\mathbf{w}_t^{\text{drift}})_i - (2\Sigma\mathbf{w}_t^{\text{drift}})_j}{2 \times 0.1425\% + \text{tax}_i}.$$

Exhibit 6 runs (29.9) at each of R3's 91 signals after the first, from the drifted previous penalized target. At $\theta = 1$ the penalized path plans 31.5% of R3's turnover, leaves 31 signals untraded and adds 4.9 basis points of in-sample volatility on average. A turnover limit is the same kind of penalty (Problem 10).

Exhibit 6: Cost-penalized R3 along its path, 2019–2026

θ	Turnover (% of $\theta = 0$)	Cost (% of $\theta = 0$)	Signals without a trade	Volatility added, mean (bp)	Largest (bp)
0	100.0	100.0	0	0.0	0.0
0.25	58.4	59.0	11	0.4	1.5
0.5	44.7	45.5	18	1.5	3.8
1	31.5	31.8	31	4.9	10.8
2	22.3	22.5	40	15.3	27.7
5	9.4	9.7	74	47.9	112.8
10	4.7	4.9	85	71.0	283.0

Note: an in-sample illustration on each window's 504 joint dates, not a rule. Turnover and cost are *planned* at each signal close and summed over the 91 signals, relative to R3 itself ($\theta = 0$). The volatility added is the target's minus R3's. R3's executed turnover and costs are LM34's. Source: as Exhibit 1.

The executed trades, a session after the signal, and their costs in the wealth recursion are LM27's and LM34's; this section plans the trade at the signal close. In Boyd et al.'s (2017) single-period method, too, each trade solves a convex problem trading off expected return, risk, transaction cost and holding cost.

導讀 **交易成本讓「不交易」成為最適**

把交易成本（transaction cost）當成懲罰函數（penalty function）放進目標函數（objective function），權重 θ 有單位：一次付清的成本，對上一年的變異數，$\theta = 2/(\delta\tau)$。把買進與賣出拆開，成本變成線性，KKT 條件說：買進的基金，梯度（gradient）落在帶狀區間的下緣，賣出的落在上緣，沒有交易的基金落在中間，這就是 "no-trade band"。θ 夠大時，最適解就是完全不交易。沿著加了成本懲罰的路徑，$\theta = 1$ 只規劃 R3 原本 31.5% 的週轉率（turnover），代價是樣本內波動率平均高出 4.9 個基點。這些都是在訊號日收盤時規劃的數字，實際執行的週轉率與成本屬於 LM34。

KNOWLEDGE CHECK 6

At $\theta = 1$ on 2025-04-30, 0050's gradient is 0.017684, while the band runs from 0.015235 (buy) to 0.019085 (sell). Why is 0050 left untouched?

ANSWER

Buying 0050 helps only if its gradient lies below the buy edge, selling only if it lies above the sell edge. Priced at the budget's λ, buying a little 0050 raises the objective at the rate $0.017684 - 0.015235$, and selling at $0.019085 - 0.017684$, so neither trade helps.

8 SOLVERS AND HOW TO CHECK THEM

- [ ] solve a problem in cvxpy with scaled inputs and set tolerances, and verify it by KKT residuals
- [ ] explain what KKT necessity needs and what a solver's status guarantees, and choose a solution method

cvxpy is a modelling language for convex problems, embedded in Python (Diamond and Boyd 2016). A problem is built from convex atoms such as `quad_form`, `norm1` and `pos`, and cvxpy refuses one whose convexity its rules cannot verify. It canonicalizes the others into the standard form of a solver it selects (Agrawal et al. 2018).

In the installed cvxpy 1.9.3 the default solver is OSQP 1.1.3 for a QP, such as R3 or (29.9), and CLARABEL 0.11.1 for an LP or a volatility limit. OSQP solves a quadratic program by operator splitting and CLARABEL by an interior-point method; neither method is derived here (Stellato et al. 2020; Goulart and Chen 2024).

Scaling. A solver stops when its residuals fall below tolerances, some of them absolute. CLARABEL also tests the gap between its objective and a lower bound from its multipliers. On daily inputs R3's optimal

variance at the first signal is 1.035×10^{-5}, so CLARABEL's default absolute gap, 10^{-8}, is about 0.1% of it. On annual inputs the variance is 2.515×10^{-3}, A times larger, and the gap is 0.0004% of it (Exhibit 7).

The condition number, 9.72 to 20.46 over the 92 windows, does not change with the units, so by (11.9) an exact solve is as accurate in either. The exact solver uses the Cholesky factor, whose condition number is $\sqrt{\kappa(\Sigma)}$ by (12.6).

Tolerances. The book sets OSQP's `eps_abs` and `eps_rel` to 10^{-10}, with up to 400,000 iterations and polishing. Polishing guesses the active constraints and re-solves the KKT equations on them, with a few refinement steps. The book sets CLARABEL's `tol_gap_abs`, `tol_gap_rel` and `tol_feas` to 10^{-12} for QPs and LPs and 10^{-10} for cone programs (Problem 11). Tightening only one of CLARABEL's two gap tolerances leaves R3 unchanged at the first signal.

cvxpy's defaults for OSQP are 10^{-5} with 10,000 iterations and polishing, in place of OSQP's own 10^{-3}, 4,000 and no polishing. CLARABEL keeps its own defaults, 10^{-8} with 200. Every printed solution comes from a fresh `Problem`, ends `optimal` and passes Example 9's checks.

Exhibit 7: R3's largest weight gap to the exact solution, by units, solver and tolerances

Units	Solver	Tolerances	First signal	Worst of 92 signals
Daily	OSQP	default	8.3×10^{-9}	1.1×10^{-8}
Daily	CLARABEL	default	8.6×10^{-4}	2.1×10^{-3}
Daily	CLARABEL	the book's	3.2×10^{-8}	3.2×10^{-7}
Annual	OSQP	default	below 10^{-15}	below 10^{-15}
Annual	CLARABEL	default	1.6×10^{-5}	3.4×10^{-5}
Annual	CLARABEL	the book's	2.0×10^{-11}	8.9×10^{-9}

Note: on each window's 504 joint dates, against `nnls` on the Cholesky factor. OSQP at the book's tolerances repeats its default rows (omitted); a gap at rounding level prints as a bound. R4's worst gap is 9.5×10^{-7} (annual, CLARABEL default); the lab prints all 16 cells. Source: as Exhibit 1.

EXAMPLE 9

Checking a solver's answer

Check R4 at the last signal by its KKT residuals, a second solver and a closed form, and read cvxpy's duals. These checks answer the case's third question.

SOLUTION

Both solvers report `optimal`. The residuals use each solver's $\mathbf{y}$ and $\lambda = 2\mathbf{y}^\top \Sigma \mathbf{y}$:

Check	OSQP	CLARABEL
Stationarity on held funds	below 10^{-15}	9.3×10^{-15}
Smallest ν	below 10^{-15}	-9.3×10^{-15}
Complementarity	below 10^{-15}	1.5×10^{-14}
Weight gap to the exact solution	below 10^{-15}	1.7×10^{-13}

Where the exact $\mathbf{y}$ has its two zeros, CLARABEL leaves 3.4×10^{-13} (00646) and 2.0×10^{-13} (00679B), so a weight below 10^{-6} prints as 0. LM28's (28.6) on the held funds {0050, 00635U, 00642U} gives R4 to below 10^{-15}.

cvxpy reports an equality's dual as $-\lambda$ in LM7's convention and the bound duals as the $\nu_i \geq 0$. R3's budget dual is −0.020396, so $\sigma_p = \sqrt{\lambda/2} = 10.10\%$; R4's dual of $\mathbf{e}^\top \mathbf{y} = 1$ is −0.4506, with bound duals 0.0234 (00646) and 0.0292 (00679B). As in LM28, check the sign of an equality's dual against a closed form.

The method follows from the problem. A closed form serves when no inequality binds, and an exact active-set solution such as `nnls` serves a small long-only QP. A general solver handles caps, costs and cones.

PITFALL **"Optimal" on badly scaled inputs**

On daily inputs CLARABEL reports `optimal` for R3 and still misses its weights by up to 2.1×10^{-3}: the residuals met the tolerances, but the printed digits are wrong. Give solvers inputs on which the objective is not small against the absolute tolerances, as annual inputs are for R3. Then set the tolerances and check the KKT residuals.

導讀 **先看狀態，再看權重**

solver 回報的 "optimal" 只表示它的殘差（residual）小於容許誤差（tolerance），而有些容許誤差是絕對值：用日資料時，R3 的目標函數（objective function）只有 1.035×10^{-5}，預設的 10^{-8} 就大約是它的千分之一，權重偏離 8.6×10^{-4}，92 個訊號中最多偏離 2.1×10^{-3}。改用年化資料，同一個容許誤差相對小得多。檢查的順序是：先看狀態，再看每條限制條件（constraint）是否滿足、KKT 殘差、第二個 solver，最後與已知的公式解（closed form）比對。cvxpy 回報的對偶變量（dual variable）就是 KKT 條件的乘數（multiplier），但等式的符號要先用公式解核對。OSQP 用 operator splitting，CLARABEL 用內點法（interior-point method），本書都不推導。

KNOWLEDGE CHECK 7

CLARABEL's default absolute tolerance is 10^{-8}. At the first signal R3's optimal variance is 1.035×10^{-5} in daily units and 2.515×10^{-3} in annual units. Why does the daily problem miss the weights by 8.6×10^{-4} and the annual one by only 1.6×10^{-5}?

ANSWER

The solver stops once its gap is below an absolute 10^{-8}, about 0.1% of the daily objective. Near the optimum a weight error of order 10^{-3} moves the variance by less than that. On annual inputs the same gap is relatively A times smaller, so the solver goes further.

9 CONVEX OPTIMIZATION IN PYTHON

- [ ] solve a problem in cvxpy with scaled inputs and set tolerances, and verify it by KKT residuals

PYTHON

```python
import cvxpy as cp
import numpy as np
from scipy.linalg import cholesky, solve_triangular
from scipy.optimize import nnls

BOOK = {"tol_gap_abs": 1e-12, "tol_gap_rel": 1e-12, "tol_feas": 1e-12}


def qp(S, a):                          # min y'Sy, a'y = 1, y >= 0 in cvxpy
    y = cp.Variable(len(a))
    con = a @ y == 1
    prob = cp.Problem(cp.Minimize(cp.quad_form(y, cp.psd_wrap(S))),
                      [con, y >= 0])
    prob.solve(solver=cp.CLARABEL, **BOOK)    # a fresh Problem
    assert prob.status == "optimal" and y.value is not None
    return y.value, -float(con.dual_value)    # cvxpy's dual is -lambda


def exact(S, a):                       # the same problem by nnls
    U = cholesky(S)                    # S = U'U, U upper triangular
    z, _ = nnls(U, solve_triangular(U, a, trans="T"))
    return z / (a @ z)


def kkt(S, a, y):                      # residuals of (29.5) and (29.6)
    lam = 2 * y @ S @ y
    nu = 2 * S @ y - lam * a
    held = y > 1e-6
    return np.abs(nu[held]).max(), nu.min(), np.abs(nu * y).max()

# R3: a = np.ones(5); R4: a = e; the weights are y / y.sum().
# nnls minimizes |U z - U^-T a|^2 over z >= 0, which is z'Sz - 2a'z plus a
# constant; then y = z / a'z. psd_wrap declares S PSD without testing it,
# so pass only a matrix known to be positive definite.
```

IN PYTHON **Companion lab**

The Polars-first notebook `LM29_lab.ipynb` rebuilds the module's exhibits and examples, R3 and R4 at all 92 signals included. It solves Practice Problems 9 to 12 and ends with an enumeration of R4's active sets.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

In cvxpy 1.9.3, which solver does `cp.Problem(cp.Maximize(mu @ w), [cp.sum(w) == 1, w >= 0, w <= 0.4]).solve()` call? Why should a script name its solver anyway? Which method suits R3 on E5, R3 with a 40% cap and (29.9)?

ANSWER

The problem is an LP, which cvxpy 1.9.3 sends to CLARABEL. A script names its solver and tolerances because defaults change between versions. A closed form serves R3 on E5, where no bound binds; the cap and (29.9)'s cost need a general solver.

SUMMARY

- A convex problem (29.3) has no false minima; test convexity by the Hessian (29.2), by sums and by maxima.
- For a convex objective with linear constraints the KKT conditions (29.4) are necessary and sufficient; long-only minimum variance holds funds of beta 1, (29.5).
- R4 is the QP (29.6); a fund is held only if its excess return covers what its beta requires, (29.7).
- A multiplier is a shadow price (29.8), whose linear prediction is the most a relaxation gains.
- Costs in the objective (29.9) create a no-trade band; $\theta = 2/(\delta\tau)$.
- Give solvers annual inputs and set tolerances; read the status, then the KKT residuals.

PRACTICE PROBLEMS

Problems on data use the through case's windows (504 joint daily total returns through the signal, $A = 242.95$, divisor T, annual inputs, $r_f = 1.5\%$); a weight below 10^{-6} is 0. Solvers run in cvxpy 1.9.3 at the book's tolerances unless stated. Costs are 0.1425% a side plus a 0.1% tax on sales (none for 00679B), planned at the signal close.

1. For a positive definite Σ, the set $\{\mathbf{w} \in \mathbb{R}^N : \mathbf{w}^\top \Sigma \mathbf{w} \geq 0.01\}$ is:

   A. convex, because Σ is positive definite

   B. convex, because its constraint is a polynomial

   C. not convex: it holds $-\mathbf{w}$ whenever it holds $\mathbf{w}$, but not their midpoint $\mathbf{0}$

2. Long-only minimum variance ends at $\sigma_p^2 = 0.0102$ (annual), and an excluded fund has $\mathrm{Cov}(R_i, R_p) = 0.015$ (hypothetical). Its multiplier ν_i in (29.5) is *closest* to:

   A. 0.0096

   B. 0.0048

C. −0.0096

3. On the last window the means are (48.775, 17.937, −5.776, 20.678, 36.096)% in the funds' usual order. The highest mean subject to the budget and $0 \le w_i \le 40\%$ comes from:

   A. 20% in each fund

   B. 40% in 0050, 40% in 00642U and 20% in 00635U

   C. 100% in 0050

4. Minimum variance of five funds with the budget and a 15% cap on each is:

   A. solved by 15% in each fund

   B. unbounded

   C. infeasible

5. cvxpy reports the dual −0.4506 of $\mathbf{e}^\top \mathbf{y} = 1$ in R4's QP (29.6) on the last window. The in-sample Sharpe ratio is *closest* to:

   A. 2.107

   B. 1.490

   C. 0.4506

6. cvxpy reports the budget dual −0.020396 for R3 on the last window. R3's in-sample volatility is *closest* to:

   A. 14.28%

   B. 10.10%

   C. undefined, because the dual is negative

7. On 2025-04-30, $2\Sigma \mathbf{w}_t^{\text{drift}} = (0.02318, 0.02513, 0.01383, 0.01280, 0.02503)$ in the funds' usual order, and every drifted weight is positive. Find the smallest θ at which (29.9) makes no trade, and the pair of funds that sets it.

8. (a) Prove the KKT conditions sufficient for $\min h(\mathbf{w})$ subject to $K\mathbf{w} = \mathbf{b}$ and $\mathbf{w} \ge \mathbf{0}$, with h convex and differentiable. (b) With the budget and caps $w_i \le b_i$, show that $\lambda = 2V + \sum_i b_i \nu_i^{\text{cap}}$ for $h = \mathbf{w}^\top \Sigma \mathbf{w}$, and check it on Example 7. (c) Prove from (29.1) that a convex problem's local minimum is global, and show by (29.2) that $-\sum_i \ln w_i$ is strictly convex on $\mathbf{w} > \mathbf{0}$.

9. First window, 2017-02-13 to 2019-02-27. (a) Why does R3 equal LM28's $\mathbf{w}^{\text{mv}}$, (10.7, 22.5, 41.4, 25.0, 0.4)%? (b) *(Python)* Compute R4 by `nnls`, its in-sample Sharpe ratio against LM28's unconstrained 0.880, and the excluded funds' ν_i/λ. (c) Check R4 by (28.6) on its held funds.

10. *(Python)* On 2025-04-30, solve R3 with $\frac{1}{2}\|\mathbf{w} - \mathbf{w}_t^{\text{drift}}\|_1 \le 0.05$ by CLARABEL: weights, volatility and the limit's multiplier. Show that the penalty $\theta\,\text{TO}_t$, with LM27's turnover TO_t and θ equal to that multiplier, gives the same weights, say why, and weigh turnover against in-sample risk.

11. *(Python)* On the last window, maximize $\mathbf{f}^\top\mathbf{e} - \text{spread} \times \max(0, \mathbf{1}^\top\mathbf{f} - 1)$ subject to $\mathbf{f}^\top\Sigma\mathbf{f} \le (25\%)^2$, $\mathbf{f} \ge \mathbf{0}$ and $\|\mathbf{f}\|_1 \le 1.5$, with the unrounded $\mathbf{e}$ and an illustrative spread over r_f of 0 and of 2 points. Solve by CLARABEL at 10^{-10} and SCS 3.3.1 (`eps` 10^{-9}, 200,000 iterations), find the exact solution, and say which limit binds.

12. *(Python)* (a) Solve R3 at the first signal in daily and annual units, by OSQP and CLARABEL at default and book tolerances; report the largest weight gap to `nnls`. (b) Repeat over the 92 signals, and for R4, whose daily form uses $\hat{\Sigma}$ and $\bar{\mathbf{R}} - r_f/A$; compare $\mathbf{w} = \mathbf{y}/\mathbf{1}^\top\mathbf{y}$. (c) Explain the pattern, and why OSQP's polishing hides it. (d) Compare (29.5)'s largest stationarity residual, as a share of λ, for CLARABEL's default R3 at the first signal in the two units.

SOLUTIONS

1. **C** is correct: $(-\mathbf{w})^\top\Sigma(-\mathbf{w}) = \mathbf{w}^\top\Sigma\mathbf{w}$, but the midpoint $\mathbf{0}$ has variance 0 (cvxpy raises a `DCPError`). A confuses the convex function with its upper level set, and a polynomial constraint (B) need not define a convex set.

2. **A** is correct: $\nu_i = 2(0.015 - 0.0102) = 0.0096$, and $\beta_{i,p} = 1.47$. B drops the 2 of the gradient $2\Sigma\mathbf{w}$. C reverses the sign, which (29.5) forbids.

3. **B** is correct: its mean is $0.4 \times 48.775 + 0.4 \times 36.096 + 0.2 \times 20.678 = 38.08\%$, optimal by Example 2. A holds low-mean funds, and C ignores the cap.

4. **C** is correct: the caps sum to 75%, so no portfolio meets the budget. A breaks the budget, and B would need feasible points.

5. **A** is correct: $\text{SR}^* = \sqrt{2/0.4506} = 2.107$ by (29.6). B drops the 2, $\sqrt{1/0.4506} = 1.490$, and C reads the dual as the ratio.

6. **B** is correct: the dual is $-\lambda$ with $\lambda = 2\sigma_p^2$, so $\sigma_p = \sqrt{0.020396/2} = 10.10\%$. A drops the 2, $\sqrt{0.020396} = 14.28\%$, and C keeps cvxpy's sign.

7. By Section 7's Deep Dive the threshold is the largest ratio in its last display. Selling 00646 against buying 00635U gives the largest ratio, $(0.02513 - 0.01280)/0.00385 = 3.2026$ (3.2022 unrounded): no trade for $\theta \ge 3.2022$.

8. (a) With $\psi_i(\mathbf{w}) = -w_i$, (29.2) and stationarity give $h(\mathbf{w}) - h(\mathbf{w}^*) \ge \boldsymbol{\lambda}^\top K(\mathbf{w} - \mathbf{w}^*) + \boldsymbol{\nu}^\top(\mathbf{w} - \mathbf{w}^*) = \boldsymbol{\nu}^\top\mathbf{w} \ge 0$, since $\boldsymbol{\nu}^\top\mathbf{w}^* = 0$. (b) Stationarity is $2\Sigma\mathbf{w} = \lambda\mathbf{1} + \boldsymbol{\nu} - \boldsymbol{\nu}^{\text{cap}}$. Multiplying by $\mathbf{w}^\top$, with $\boldsymbol{\nu}^\top\mathbf{w} = 0$ and $\nu_i^{\text{cap}} w_i = \nu_i^{\text{cap}} b_i$, gives $2V = \lambda - \sum_i b_i \nu_i^{\text{cap}}$; on Example 7, $2 \times 0.0110665 + 0.40 \times 0.010543 = 0.026350$. (c) A feasible point of lower value would give, by (29.1), lower values on its segment arbitrarily close to the local minimum. The Hessian of $-\sum_i \ln w_i$ is $\text{diag}(1/w_1^2, \ldots, 1/w_N^2)$, PD on the open convex set $\mathbf{w} > \mathbf{0}$.

9. (a) Every weight of $\mathbf{w}^{\text{mv}}$ is positive, and it is the minimum over all fully invested portfolios, hence over the long-only ones; (29.5) holds with $\boldsymbol{\nu} = \mathbf{0}$. (b) R4 holds 10.57% in 0050, 52.44% in 00646 and

36.99% in 00679B, with a Sharpe ratio of 0.802 against 0.880; ν_i/λ is 0.25 points for 00635U and 8.01 for 00642U. (c) (28.6) on {0050, 00646, 00679B} gives the same weights to rounding level.

10. The limit gives (2.2, 27.4, 36.9, 27.7, 5.8)%, a volatility of 9.24% and a multiplier of 0.00825. The penalty with θ equal to the unrounded multiplier gives the same weights within 10^{-9}. Its stationarity is the limit's, with θ for the multiplier, and both problems are strictly convex.

    The limit costs risk: 9.24% against R3's 8.97% at a turnover of 18.72%, and no trade's 9.51%. It trades 27% of R3's turnover and closes about half of the volatility gap between no trade and R3.

11. Since $\max(0, L-1) \geq L-1$ for $L = \mathbf{1}^\top \mathbf{f}$, the objective is at most $(\mathbf{e} - \text{spread}\,\mathbf{1})^\top \mathbf{f} + \text{spread}$, with equality when $L \geq 1$. Over the limits, that bound is largest at the long-only tangency at r_f plus the spread, scaled to 25% volatility. That tangency's volatility is below 25%, so $L > 1$ and the bound is attained, with gross exposure below 150%. With no spread that tangency is R4, scaled by $25/18.85 = 1.326$; with 2 points it is the tangency at 3.5%, scaled by $25/19.08 = 1.310$. 00646 and 00679B get 0:

	0050	00635U	00642U	L (%)	Excess return (%)
R4	62.77	18.37	18.86		
$\mathbf{f}$, no spread	83.23	24.36	25.00	132.6	52.67
Tangency at 3.5%	64.40	16.67	18.93		
$\mathbf{f}$, 2 points	84.36	21.83	24.81	131.0	52.65

    After financing, the 2-point excess return is 52.03%, $2 \times 0.310 = 0.62$ points less. CLARABEL and SCS agree within 10^{-6}; at 10^{-12} CLARABEL returns `optimal_inaccurate`. Each exposure is a scale times an unrounded weight, so its last digit can differ from the printed product. The excess returns recompute from the exposures and Exhibit 2's three-decimal e_i. At r_f of 0% and 2% the 2-point optimum has $L = 132.2\%$ and 130.6%, and 54.01% and 51.38% after financing.

12. (a) and (b) are Exhibit 7; the lab prints all 16 cells. (c) CLARABEL's 10^{-8} gap is absolute, and R3's daily objective, 1.035×10^{-5}, is A times smaller than the annual 2.515×10^{-3}. R4's constraint $\mathbf{e}^\top \mathbf{y} = 1$ makes its daily $\mathbf{y}$, and so its objective $1/\text{SR}^{*2}$, A times larger: 378 against 1.556 at the first signal. Both exceed 1, and R4's worst default gaps, 3.3×10^{-7} daily and 9.5×10^{-7} annual, show no such pattern.

    Once OSQP's polishing (Section 8) has the right active set, its answer no longer depends on `eps_abs` and `eps_rel`, so R3's default and book rows agree. Its refinement still leaves 8.3×10^{-9} on daily inputs.

    (d) Every fund is held, so each $(2\Sigma\mathbf{w})_i$ should equal λ. The largest gap is 1.6% of λ on daily inputs and 0.03% on annual ones: the residuals expose the daily solution.

GLOSSARY

Term	中文	Meaning
Convex set	凸集	A set containing the segment between any two of its points, (29.1).
Convex function	凸函數	A function whose chords lie on or above its graph, (29.1).
Strictly convex function	嚴格凸函數	A convex function whose chords lie strictly above the graph between distinct points.
Convex combination	凸組合	$\omega\mathbf{u} + (1-\omega)\mathbf{v}$ with $\omega \in [0,1]$; more generally, a mix with non-negative weights summing to 1.
Convex program	凸規劃	Minimizing a convex function subject to convex $\psi_i \le b_i$ and $K\mathbf{w} = \mathbf{b}$, (29.3).
Quadratic program	二次規劃	A convex quadratic objective with linear constraints.
Linear program	線性規劃	A linear objective with linear constraints.
Feasible set	可行解區域	The points that satisfy every constraint.
KKT conditions	KKT 條件	Stationarity, feasibility, non-negative multipliers and complementary slackness, (29.4).
Complementary slackness	互補鬆弛性	Each inequality binds or has a zero multiplier.
Constraint qualification	約束規範	A condition on the constraints under which every minimum is a KKT point; linear constraints satisfy one.
Active constraint	有效限制	An inequality that holds with equality at the optimum; the active set lists them.
Shadow price	影子價格	The rate $\nu_i = -\partial J/\partial b_i$ at which relaxing a limit improves the optimal value, (29.8).
Homogenized QP	—	R4 as $\min \mathbf{y}^\top \Sigma \mathbf{y}$ subject to $\mathbf{e}^\top \mathbf{y} = 1$, $\mathbf{y} \ge \mathbf{0}$, (29.6).
Penalty function	懲罰函數	A term added to the objective in place of a limit, such as θ cost in (29.9).
No-trade band	—	The range of a fund's variance gradient in which a cost-penalized trade leaves the fund untouched.
Dual variable	對偶變量	A solver's multiplier of a constraint; cvxpy's `dual_value`.
Interior-point method	內點法	A method that approaches a KKT point while it keeps every inequality's multiplier strictly positive; its iterates need not be feasible (CLARABEL).
Solver status	—	The solver's verdict (`optimal`, `optimal_inaccurate`, `infeasible`, `unbounded`), read before its answer.

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